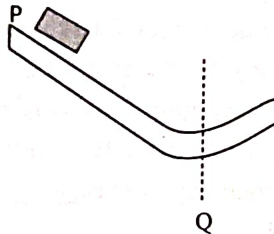
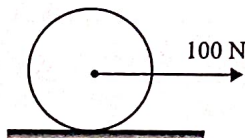


50. Block P of mass 2 kg slides down the surface and has a speed 20 m/s at the lower point Q, where the local radius of curvature is 2 m as shown in the figure. Assuming $g = 10 \text{ m/s}^2$, the normal force (in N) at Q is _____ (correct to two decimal places).



(GATE - 18 - SET - 1)

51. A force of 100 N is applied to the centre of a circular disc, of mass 10 kg and radius 1 m, resting on a floor as shown in the figure. If the disc rolls without slipping on the floor, the linear acceleration (in m/s^2) of the centre of the disc is _____ (correct to two decimal places).

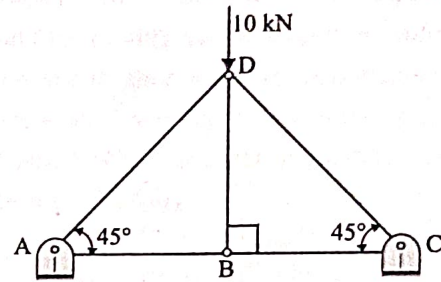


(GATE - 18 - SET - 2)

52. A rigid link PQ of length 1.0 m is pinned at P. It rotates about P in a vertical plane with a uniform angular acceleration of 1.0 rad/s^2 . At an instant when the angular velocity of the link is 1.0 rad/s , the magnitude of total acceleration (in m/s^2) of point Q relative to point P is _____ (PI_GATE - 18)

- (a) 1.41 (b) 1.73
(c) 2 (d) 2.83

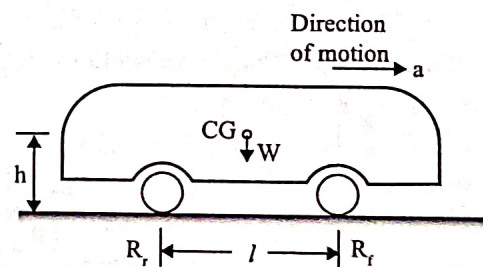
53. A truss is composed of members AB, BC, CD, AD and BD, as shown in the figure. A vertical load of 10 kN is applied at point D. The magnitude of force (in kN) in the member BC is _____.



(GATE - 19 - SET - 1)

54. A car having weight W is moving in the direction as shown in the figure. The center of gravity (CG) of the car is located at height h from the ground, midway between the front and rear wheels. The distance between the front and rear wheels is l . The acceleration of the car is a , and acceleration due to gravity is g . The reactions on the front wheels (R_f) and rear wheels (R_r) are given by

(GATE - 19 - SET - 1)

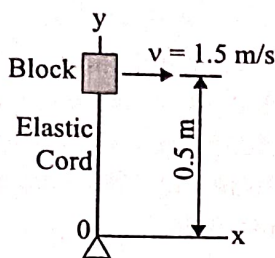


- (a) $R_f = R_r = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{l} \right) a$
(b) $R_f = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{l} \right) a$; $R_r = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{l} \right) a$
(c) $R_f = R_r = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{l} \right) a$
(d) $R_f = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{l} \right) a$; $R_r = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{l} \right) a$

55. A ball of mass 3 kg moving with a velocity of 4 m/s undergoes a perfectly-elastic direct-central impact with a stationary ball of mass m . After the impact is over, the kinetic energy of the 3 kg ball is 6 J. The possible value(s) of m is/are (GATE - 19 SET - 2)

- (a) 6 kg only (b) 1 kg only
(c) 1 kg, 6 kg (d) 1 kg, 9 kg

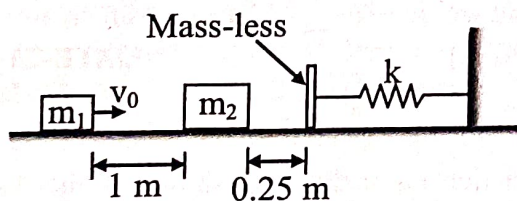
56. The 2 kg block shown in figure (top view) rests on a smooth horizontal surface and is attached to a massless elastic cord that has a stiffness 5 N/m.



The cord hinged at O is initially unstretched and always remains elastic. The block is given as velocity v of 1.5 m/s perpendicular to the cord. The magnitude of velocity in m/s of the block at the instant the cord is stretched by 0.4 m is

(GATE-20-SET-1)

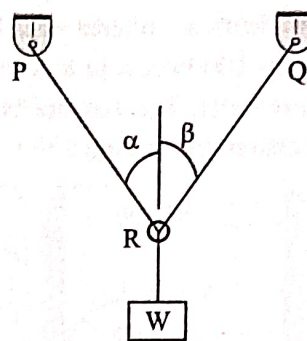
- (a) 1.07 (b) 1.36 (c) 1.50 (d) 0.83
57. A rigid block of mass $m_1 = 10$ kg having velocity $v_0 = 2$ m/s strikes a stationary block of mass $m_2 = 30$ kg after traveling 1 m along a frictionless horizontal surface as shown in the figure.



The two masses stick together and jointly move by a distance of 0.25 m further along the same frictionless surface, before they touch the mass-less buffer that is connected to the rigid vertical wall by means of a linear spring having a spring constant $k = 10^5$ N/m. The maximum deflection of the spring is _____ cm (Round off to 2 decimal places).

(GATE-20-SET-2)

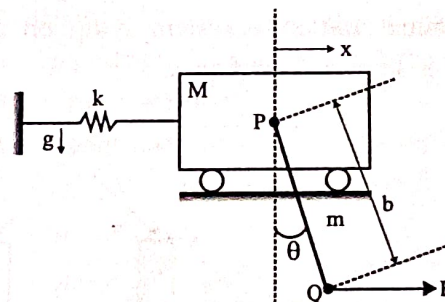
58. A truss with two bars PR and QR, making angles α and β , respectively, with the vertical, is shown in the figure below. The connections at P, Q and R are hinged connections. The truss supports a body of weight W (in N) at R as shown. The tension in the bar QR (in N) is



(GATE-PI-20)

- (a) $\frac{W \cos \beta}{\sin(\alpha + \beta)}$ (b) $\frac{W \cos \alpha}{\cos(\alpha + \beta)}$
(c) $\frac{W \sin \alpha}{\sin(\alpha + \beta)}$ (d) $\frac{W \sin \beta}{\cos(\alpha + \beta)}$

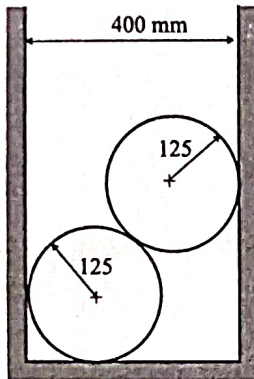
59. Consider a two degree of freedom system as shown in the figure, where PQ is a rigid uniform rod of length, b and mass, m .



Assume that the spring deflects only horizontally and force F is applied horizontally at Q. For this system, the Lagrangian, L is (GATE-21_SET-1)

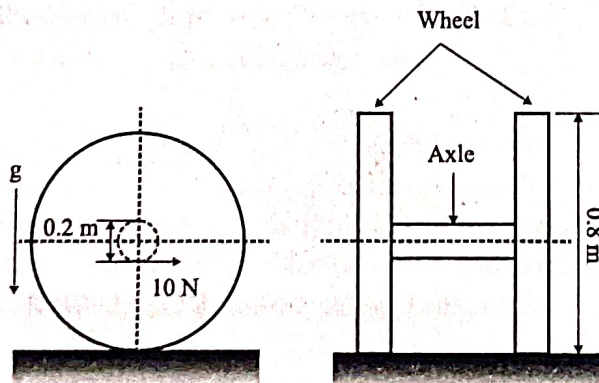
- (a) $\frac{1}{2} M \dot{x}^2 + \frac{1}{2} m b \dot{\theta} \dot{x} \cos \theta + \frac{1}{6} m b^2 \dot{\theta}^2 - \frac{1}{2} k x^2$
(b) $\frac{1}{2} M \dot{x}^2 + \frac{1}{2} m b \dot{\theta} \dot{x} \cos \theta + \frac{1}{6} m b^2 \dot{\theta}^2 - \frac{1}{2} k x^2$
 $+ m g \frac{b}{2} \cos \theta + F b \sin \theta$
(c) $\frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{2} m b \dot{\theta} \dot{x} \cos \theta + \frac{1}{6} m b^2 \dot{\theta}^2$
 $- \frac{1}{2} k x^2 + m g \frac{b}{2} \cos \theta$
(d) $\frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{6} m b^2 \dot{\theta}^2 - \frac{1}{2} k x^2 + m g \frac{b}{2} \cos \theta$

60. Two smooth identical spheres each of radius 125 mm and weight 100 N rest in a horizontal channel having vertical walls. The distance between vertical walls of the channel is 400 mm.



The reaction at the point of contact between two spheres is ____ N (round off to one decimal place)
(GATE-21_SET-1)

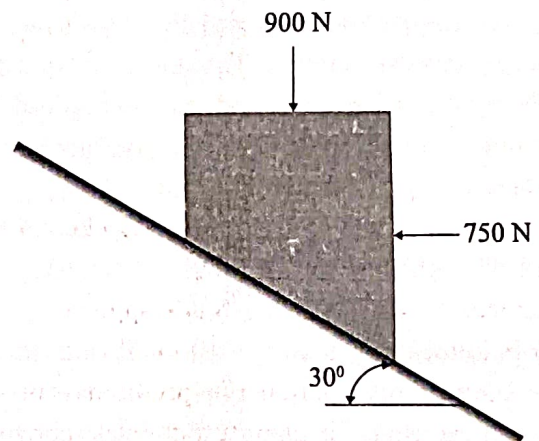
61. The wheels and axle system lying on a rough surface is shown in the figure.



Each wheel has diameter 0.8 m and mass 1 kg. Assume that the mass of the wheel is concentrated at rim and neglect the mass of the spokes. The diameter of axle is 0.2 m and its mass is 1.5 kg. Neglect the moment of inertia of the axle and assume $g = 9.8 \text{ m/s}^2$. An effort of 10 N is applied on the axle in the horizontal direction shown at mid span of the axle. Assume that the wheels move on a horizontal surface without slip. The acceleration

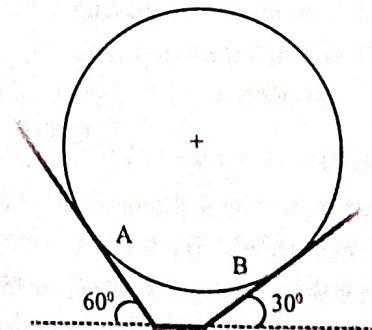
of the wheel axle system in horizontal direction is ____ m/s^2 (round off to one decimal place).
(GATE-21_SET-2)

62. A block of negligible mass rests on a surface that is inclined at 30° to the horizontal plane as shown in the figure. When a vertical force of 900 N and a horizontal force of 750 N are applied, the block is just about to slide.



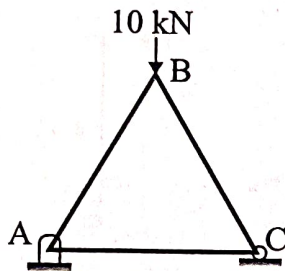
The coefficient of static friction between the block and surface is ____ (round off to two decimal places).
(GATE-21_SET-2)

63. A 30 kg smooth, solid sphere rests on two frictionless inclines as shown in the figure. The magnitude of contact force in N acting at the point A is (take acceleration due to gravity $g = 9.81 \text{ m/s}^2$ and consider both sphere and inclines to be rigid) ____ [round off to 2 decimal places]



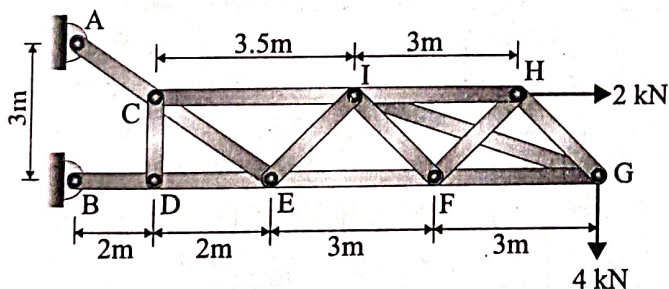
(GATE_PI-21)

64. Consider the truss shown in the figure. The members AB, BC, and CA are all rigid and form an equilateral triangle. The contact between roller and ground at C is frictionless. If the self-weight of members is neglected, the force in member BC in N is (negative sign should be used if the force is compressive and positive if the force in the member is tensile) _____ [round off to one decimal place]



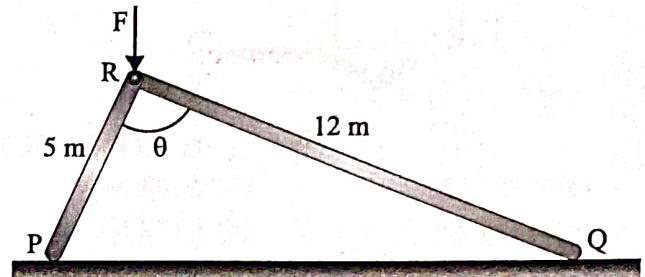
(GATE_PI-21)

65. A structure, along with the loads applied on it, is shown in the figure. Self-weight of all the members is negligible and all the pin joints are frictionless. AE is a single member that contains pin C. Likewise, BE is a single member that contains pin D. Members GI and FH are overlapping rigid members. The magnitude of the force carried by member CI is _____ kN (in integer).



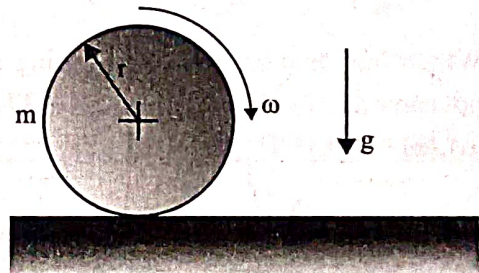
(GATE-22_SET-1)

- impending slip occurs simultaneously at points P and Q. Then the ratio of the coefficient of friction at Q to that at P (μ_Q/μ_P) is _____ (round off to two decimal places).



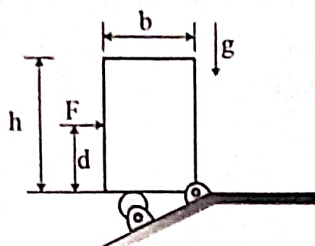
(GATE-22_SET-1)

67. A cylindrical disc of mass $m = 1$ kg and radius $r = 0.15$ m was spinning at $\omega = 5$ rad/s when it was placed on a flat horizontal surface and released (refer to the figure). Gravity g acts vertically downwards as shown in the figure. The coefficient of friction between the disc and the surface is finite and positive. Disregarding any other dissipation except that due to friction between the disc and the surface, the horizontal velocity of the center of the disc, when it starts rolling without slipping, will be _____ m/s (round off to 2 decimal places).



(GATE-22_SET-1)

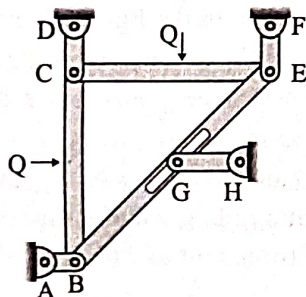
68. A rigid homogeneous uniform block of mass 1 kg, height $h = 0.4$ m and width $b = 0.3$ m is pinned at one corner and placed upright in a uniform gravitational field ($g = 9.81$ m/s²), supported by a roller in the configuration shown in the figure. A short duration (impulsive) force F , producing an impulse I_F is applied at a height of $d = 0.3$ m from the bottom as shown. Assume all joints to be frictionless. The minimum value of I_F required to topple the block is



(GATE-22_SET-2)

- (a) 0.953 Ns (b) 1.403 Ns
(c) 0.814 Ns (d) 1.172 Ns

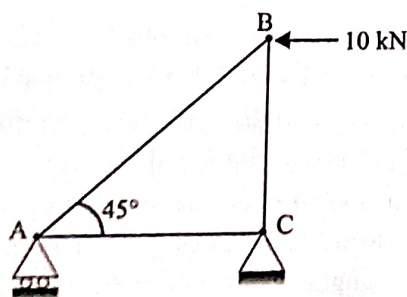
69. The lengths of members BC and CE in the frame shown in the figure are equal. All the members are rigid and lightweight, and the friction at the joints is negligible. Two forces of magnitude $Q > 0$ are applied as shown, each at the mid-length of the respective member on which it acts.



Which one or more of the following members do not carry any load (force)? (GATE-22_SET-2)

- (a) AB (b) CD (c) EF (d) GH

70. In the three-member truss shown in the figure, $AC = BC$. An external force of 10 kN is applied at B, parallel to AC. The force in the member BC is

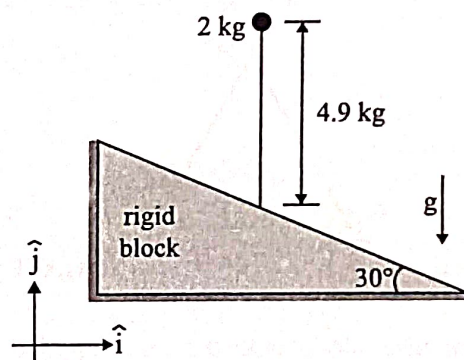


(GATE-PI-22)

- (a) zero (b) 10 kN (tensile)
(c) 10 kN (compressive) (d) 7.07 kN (tensile)

71. A spherical ball weighing 2 kg is dropped from a height of 4.9 m onto an immovable rigid block as shown in the figure. If the collision is perfectly elastic, what is the momentum vector of the ball (in kg m/s) just after impact?

Take the acceleration due to gravity to be $g = 9.8 \text{ m/s}^2$. Options have been rounded off to one decimal place. (GATE_ME-23)



- (a) $19.6 \hat{i}$ (b) $19.6 \hat{j}$
(c) $17.0 \hat{i} + 9.8 \hat{j}$ (d) $9.8 \hat{i} + 17.0 \hat{j}$

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (c)	02. (b)	03. (b)	04. (d)	05. (b)
06. (d)	07. (d)	08. (a)	09. (b)	10. (a)
11. (a)	12. (a)	13. (d)	14. (c)	15. (a)
16. (b)	17. (c)	18. (a)	19. (b)	20. (a)
21. 20	22. (d)	23. (c)	24. 100	25. 8
26. (a)	27. 57.73	28. (c)	29. (d)	30. (a)
31. (c)	32. 10	33. (b)	34. 2.414	35. (c)
36. 10	37. (d)	38. 0	39. (a)	40. (d)
41. (d)	42. 1500	43. (a)	44. (d)	45. (a)
46. 10	47. (b,c,d)	48. 3	49. 0.34	

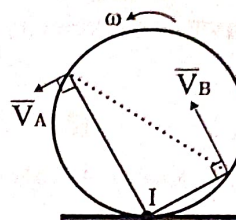
TWO MARKS QUESTIONS

01. (c)	02. (a)	03. (a)	04. a-4,b-3,c-1,d-5	
05. (d)	06. (d)	07. (a)	08. (c)	09. (d)
10. (b)	11. (b)	12. (c)	13. (c)	14. (b)
15. (a)	16. (a)	17. (a)	18. (a)	19. (b)
20. (b)	21. (b)	22. (a)	23. (b)	24. (b)
25. (d)	26. 4.905	27. 1.198	28. 40	29. 57.67
30. (c)	31. 10	32. (a)	33. 2.053	34. (a)
35. 400	36. (c)	37. 0.64	38. 29.63	39. (c)
40. 251.32	41. 133.33	42. 5	43. (b,d)	
45. 20	46. 3	47. 680	48. (a)	49. 7.5
50. 420	51. 6.66	52. (a)	53. 5	54. (b)
55. (d)	56. (b)	57. (1)	58. (c)	59. (c)
60. 125	61. 1.4	62. 0.17	63. 147.15	
64. 5773.5	65. 18	66. 5.76	67. 0.25	68. (a)
69. (b,d)	70. (b)	71. (c)		

One Mark Solutions

01. Ans: (c)

Sol:



02. Ans: (b)

Sol: Instantaneous center of a body rolling with sliding on a stationary curve is always on the common normal at the point of contact

03. Ans: (b)

Sol: The frictional force always acts in the opposite direction of motion.

In this case the acceleration at the point contact will pass through the geometric center.

04. Ans: (d)

Sol:

Tension,

$$T = m/\omega^2 + mg \cos \theta$$

For $T_{\max}$,

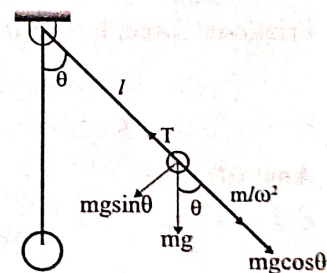
$\theta = 0^\circ$ i.e., vertical

Tension at bottom

$$T = m/\omega^2 + mg$$

Tension at top

$$T_1 = m/\omega^2 - mg$$



05. Ans: (b)

Sol: $m_A = m_B = m$ (given)

Velocity of ball 'A' before impact

$$u_A = \sqrt{2gh}$$

$$u_B = 0$$

$$\text{Velocity of approach} = u_A - u_B = \sqrt{2gh}$$

By applying conservation of linear momentum,

$$m_A u_A + 0 = m_A V_A + m_B V_B$$

$$\Rightarrow u_A = V_A + V_B \text{ ----- (1)}$$

For perfect elastic collision, $e = 1$

$\therefore$ Velocity of approach = velocity of separation

$$\Rightarrow u_A - u_B = V_B - V_A$$

$$\Rightarrow u_A = V_B - V_A \text{ ----- (2)}$$

$$V_A + V_B = V_B - V_A$$

$$\Rightarrow 2V_A = 0 \Rightarrow V_A = 0$$

$$\therefore u_A = V_B = \sqrt{2gh}$$

06. Ans: (d)

Sol: F = Friction force

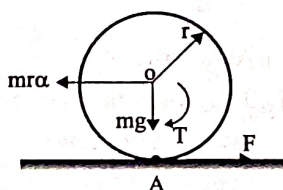
$$I_o = mr^2$$

$$\Sigma M_o = 0$$

$$T - F \times r = I_o \alpha$$

$$\Rightarrow T - F \times r = mr^2 \alpha$$

$$F = mr\alpha$$



$$\therefore T - mr^2 \alpha = mr^2 \alpha$$

$$\Rightarrow T = mr^2 \alpha + mr^2 \alpha = 2mr^2 \alpha$$

$$\therefore \alpha = \frac{T}{2mr^2}$$

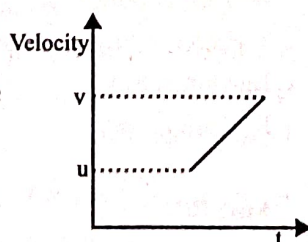
Frictional force, $F = mr\alpha$

$$= mr \times \frac{T}{2mr^2} = \frac{T}{2r}$$

07. Ans: (d)

$$\text{Sol: } S = ut + \frac{1}{2}at^2$$

$$\Rightarrow 450 = 5u + \frac{1}{2}a \times 5^2$$



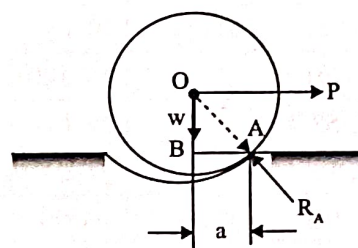
$$\text{again, } 700 = 5v + \frac{1}{2}a \times 5^2$$

Solving these equations, $v - u = 50$

$$\Rightarrow a = \frac{v - u}{t} = 10 \text{ m/sec}^2$$

08. Ans: (a)

Sol: Rolling resistance occurs because of the deformation of the surface under a rolling load. A wheel of weight W and radius r is being pulled out of the depression and over the point A by the horizontal force P . Naturally this is a continuous process as the wheel rolls.



$$\Sigma M_A = 0 = W \times a - P \times (OB)$$

Since the depression is actually very small, the distance (OB) may be replaced by r ;

Hence, $P \times r = W \times a$

Here, the horizontal component of the surface reaction R is equal to P by inspection and is called rolling resistance.

The distance 'a' is called coefficient of rolling resistance and is expressed in mm.

$$W = 500 \text{ N,}$$

$$a = 0.3,$$

$$r = 300 \text{ mm}$$

$$P = \frac{500 \times 0.3}{300} = 0.5 \text{ N}$$

09. Ans: (b)

$$\text{Sol: } \frac{T_1}{T_2} = e^{\mu \theta}$$

10. Ans: (a)

$$\text{Sol: } \Sigma F_y = 0 \Rightarrow R_A + R_B = P \times L$$

$$\Sigma M_B = 0 \Rightarrow R_A \times 3L = PL \times \frac{3L}{2}$$

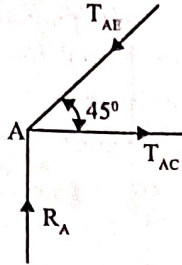
$$\Rightarrow R_A = \frac{PL}{2}, R_B = \frac{PL}{2}$$

FBD at Point A:

$$\sum F_y = 0$$

$$\Rightarrow T_{AB} \sin 45 = R_A = \frac{PL}{2}$$

$$\Rightarrow T_{AE} = \frac{PL}{\sqrt{2}}$$



$$\sum F_x = 0$$

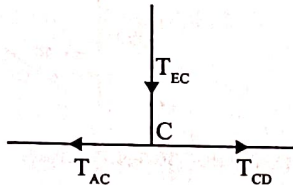
$$\Rightarrow T_{AC} = T_{AE} \cos 45 = \frac{PL}{2}$$

FBD at Point C:

$$\sum F_y = 0$$

$$\Rightarrow T_{EC} = 0$$

$$T_{AC} = T_{CD} = \frac{PL}{2}$$



11. Ans: (a)

Sol: K. E lost by the block with bullet = work done to overcome the frictional force

$$\Rightarrow (M+m) \frac{u^2}{2} = F \times S$$

Where F = Frictional force = μR

$$\Rightarrow (M+m) \frac{u^2}{2} = \mu RS$$

Again $R = (M+m)g$

$$\Rightarrow (M+m) \frac{u^2}{2} = \mu (M+m) g S$$

$$\Rightarrow u^2 = 2\mu g S$$

$$\Rightarrow u = \sqrt{2\mu g S} \text{ -----(1)}$$

Again

$$(M+m)u = mV$$

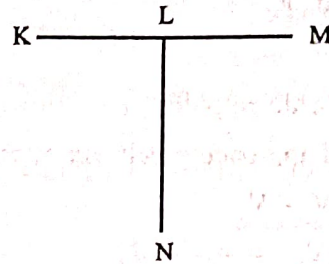
$$u = \frac{mV}{M+m} \text{ -----(2)}$$

So solving this $\frac{mV}{M+m} = \sqrt{2\mu g S}$

$$V = \frac{M+m}{m} \sqrt{2\mu g S}$$

12. Ans: (a)

Sol:



Considering joint L as shown in the figure above $\Rightarrow$

$$\sum Y = 0$$

$\Rightarrow$ The force in member LN = 0

13. Ans: (d)

$$\text{Sol: } x = 2t^3 + t^2 + 2t$$

$$V = \frac{dx}{dt} = 6t^2 + 2t + 2$$

$$a = \frac{dv}{dt} = 12t + 2$$

$$\text{At, } t = 0$$

$$\Rightarrow V = 2 \text{ and } a = 2$$

14. Ans: (c)

Sol: F.B.D of bob is shown below:



Net force at mean position = $T - mg$

Now

$$T - mg = m\omega^2 r$$

$$F_{\text{net}} = m \times \left(\frac{v}{r}\right)^2 \times r = 1 \times \left(\frac{5}{5}\right)^2 \times 5 = 5 \text{ N}$$

15. Ans: (a)

Sol: During inelastic collision only linear momentum is conserved. Both linear momentum and kinetic energy is conserved in an elastic collision.

16. Ans: (b)

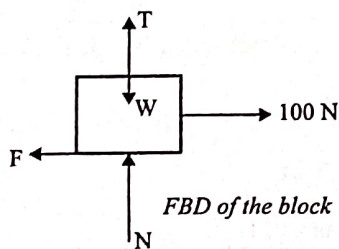
Sol: $V_r = \frac{dL}{dt} = 1 \text{ m/s}$

$$V_0 = \frac{L d\theta}{dt} = 1 \times 1 = 1 \text{ m/s}$$

$$V = \sqrt{V_r^2 + V_0^2} = \sqrt{1^2 + 1^2} = \sqrt{2} \text{ m/s}$$

17. Ans: (c)

Sol: The FBD of the above block shown



$$\Sigma Y = 0 \Rightarrow N + T - W = 0$$

$$N = W - T = 981 - T$$

$$F = \mu N = 0.2 (981 - T)$$

$$\Sigma X = 0 \Rightarrow 100 - F = 0$$

$$F = 100 = 0.2 (981 - T)$$

$$\Rightarrow T = 481 \text{ N}$$

18. Ans: (a)

Sol: During the perfectly plastic impact the colliding bodies stick together after the collision and hence velocity of separation is zero.

$$\text{Now, } e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = 0$$

19. Ans: (b)

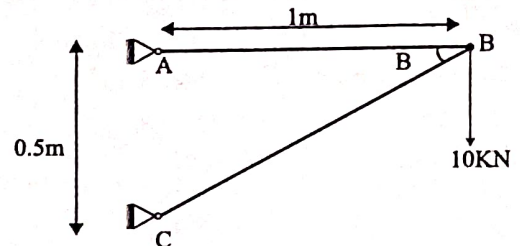
Sol: $E = \int_0^{0.1} F dx = \left[\frac{300x^3}{3} \right]_0^{0.1} = 100 \times \frac{1}{1000} = 0.1 \text{ J}$

20. Ans: (a)

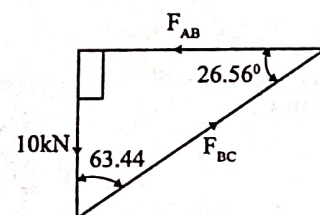
Sol: Instantaneous centre will have zero velocity because the instantaneous centre is the point of contact between the object and the floor.

21. Ans: 20 kN

Sol:



Lami's triangle :



$$\tan \theta = \frac{0.5}{1.0} \Rightarrow \theta = \tan^{-1} \left(\frac{0.5}{1} \right) = 26.56^\circ$$

From the Lami's triangle

$$\frac{10}{\sin 26.56^\circ} = \frac{F_{BC}}{\sin 90^\circ} = \frac{F_{AB}}{\sin 63.44^\circ}$$

$$F_{AB} = \frac{10}{\sin 26.56^\circ} \times \sin 63.44^\circ = 20 \text{ kN}$$

$$F_{BC} = \frac{10}{\sin 26.56^\circ} \times \sin 90^\circ = 22.36 \text{ kN}$$

22. Ans: (d)

Sol: Relative velocity after impact = 0

Coefficient of restitution

$$= \frac{\text{Relative velocity after impact}}{\text{Relative velocity before impact}} = 0$$

23. Ans: (c)

Sol: For a statically determinate plane truss,

$$m = 2j - 3$$

24. Ans: 100

Sol: $\Sigma F_y = 0$

$$F_{AB} = F_{BC} = F$$

$$2F_{AB} \times \sin \theta = 100 \text{ N}$$

$$2F_{AB} \times \sin 30^\circ = 100$$

$$F = F_{AB} = F_{BC} = \frac{100}{2 \times \frac{1}{2}}$$

$$F = F_{AB} = F_{BC} = 100 \text{ N}$$

25. Ans: 7.8 to 8.2

Sol:

$$m_1 = 1 \text{ kg} \quad m_2 = 2 \text{ kg} \quad m_1 = 1 \text{ kg} \quad m_2 = 2 \text{ kg}$$

$$u_1 = 12 \text{ m/s} \quad u_2 = 0 \quad v_1 \quad v_2$$

From momentum conservation equation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$1 \times 12 = v_1 + 2v_2 \text{-----(1)}$$

From kinetic energy conservation equation

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\frac{1}{2} \times 1 \times 12^2 = \frac{1}{2} \times 1 \times v_1^2 + \frac{1}{2} \times 2 \times v_2^2$$

$$v_1^2 + 2v_2^2 = 144 \text{-----(2)}$$

From equation (1) $v_1 = 12 - 2v_2$ substituting in equation (2)

$$(12 - 2v_2)^2 + 2v_2^2 = 144$$

$$144 + 4v_2^2 - 2 \times 2 \times 12 \times v_2 + 2v_2^2 = 144$$

$$6v_2^2 - 48v_2 = 0 \Rightarrow v_2(6v_2 - 48) = 0$$

$$\Rightarrow v_2 = \frac{48}{6} = 8 \text{ m/s}$$

26. Ans: (a)

Sol: Adopting Lami's Theorem

$$\frac{500}{\sin 90} = \frac{T_1}{\sin 120} = \frac{T_2}{\sin 150}$$

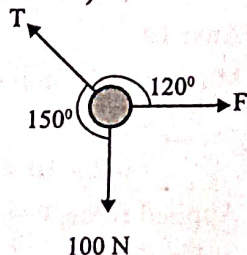
$$T_1 = 433 \text{ N}; \quad T_2 = 250 \text{ N}$$

27. Ans: 57.735N (Range: 55 to 60)

Sol: By Lami's theorem

$$\frac{F}{\sin 150} = \frac{100}{\sin 120}$$

$$\Rightarrow F = 57.735 \text{ N}$$



28. Ans: (c)

Sol: By Energy Conservation

$$mgH = \frac{1}{2} mv^2$$

$$\Rightarrow v = \sqrt{2gH}$$

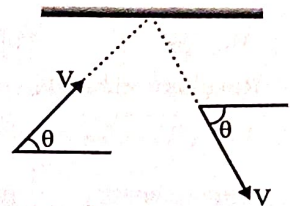
29. Ans: (d)

Sol:

$$\vec{U} = V \cos \theta \hat{i} + V \sin \theta \hat{j}$$

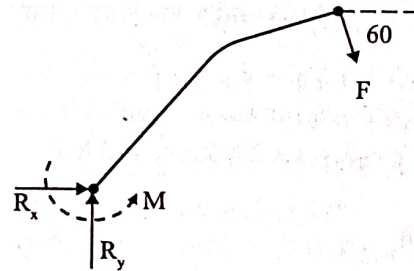
$$\vec{V} = V \cos \theta \hat{i} - V \sin \theta \hat{j}$$

$$\Delta \vec{mv} = m(\vec{V} - \vec{U}) = -2mV \sin \theta \hat{j}$$



30. Ans: (a)

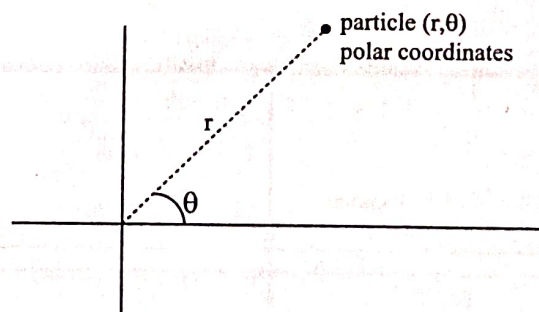
Sol: The F.B.D of bent bar



Hence, correct F.B.D is option (a).

31. Ans: (c)

Sol:



$$r(t = 2 \text{ sec}) = t^2; \quad V = \frac{dr}{dt} = 2t;$$

$$V(t = 2) = 2 \times 2 = 4 \text{ m/s};$$

$$r = 2 \times 2 = 4 \text{ m}$$

$$\theta(t) = t; \quad \omega = \frac{d\theta}{dt} = 1 \text{ rad/s};$$



$$I = mr^2 = 1 \times 4^2 = 16 \text{ kg-m}^2$$

$$\begin{aligned} \text{Total kinetic energy} &= \frac{1}{2} mV^2 + \frac{1}{2} I\omega^2 \\ &= \frac{1}{2} \times 1 \times 4^2 + \frac{1}{2} \times 16 \times 1^2 \\ &= 8 + 8 = 16 \text{ N-m} \end{aligned}$$

Alternate Solution:

$$V_r = \frac{dr}{dt} = 2t = 4, V_\theta = r \frac{d\theta}{dt} = r = t^2 = 4$$

Resultant velocity,

$$V = \sqrt{V_r^2 + V_\theta^2} = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

$$\begin{aligned} \text{Kinetic energy} &= \frac{1}{2} mV^2 = \frac{1}{2} \times 1 \times (4\sqrt{2})^2 \\ &= 16 \text{ N-m} \end{aligned}$$

32. Ans: 10

Sol: Total distance covered

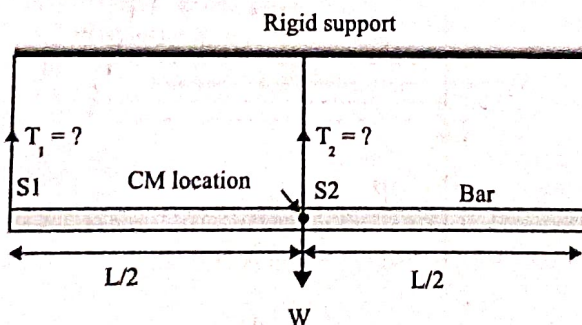
= Area under velocity – time graph

$$\begin{aligned} &= \frac{1}{2} \times 1 \times 1 + 1 \times 1 + 1 \times 1 \\ &\quad + \frac{1}{2} \times 3 \times 1 + 2 \times 2 + \frac{1}{2} \times 2 \times 2 \\ &= 10 \text{ m} \end{aligned}$$

33. Ans: (b)

Sol: FBD of the bar is shown in figure.

Balancing moments about B,



$$T_1 \times \frac{L}{2} = 0 \Rightarrow T_1 = 0$$

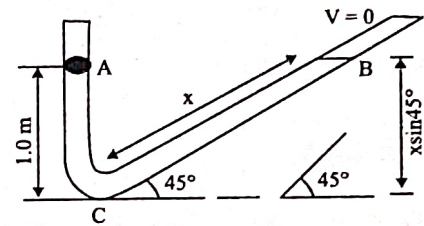
Now, balancing forces, $T_1 + T_2 = 100 \text{ N}$

$$\Rightarrow 0 + T_2 = 100 \text{ N}$$

$$\Rightarrow T_2 = 100 \text{ N}$$

34. Ans: 2.414

Sol:



Let, A = point of release,

B = point at which body comes to rest

Applying law of conservation of energy, (since no friction)

$$(K.E + P.E)_A = (K.E + P.E)_B$$

$$0 + P.E_A = P.E_B + 0$$

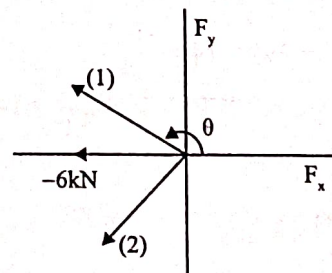
$$mgh = mgx \sin 45^\circ$$

$$h = \frac{x}{\sqrt{2}}, x = \sqrt{2} \times h = \sqrt{2} \text{ m}$$

$$\text{Total distance traveled} = 1 + \sqrt{2} = 2.414 \text{ m}$$

35. Ans: (c)

Sol:



$$F = 10 \text{ kN}$$

$$F_x = F \cos \theta$$

$$\therefore 10 \cos \theta = -6 \text{ kN}$$

$$\therefore \theta = 126.86^\circ \text{ [IIQ, IIIQ]}$$

$$\begin{aligned} \therefore F_y &= 10 \sin \theta = 8 \text{ kN (IIQ)} \\ &= -8 \text{ kN (IIIQ)} \end{aligned}$$

36. Ans: 10

Sol: Maximum friction force,

$$f_{\max} = \mu N$$

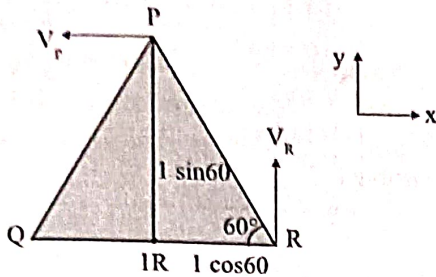
$$= 0.2 \times 10 \times 9.81 = 19.62 \text{ N}$$

$$\text{Applied force, } P = 10 \text{ N} < f_{\max}$$

$$\therefore \text{Friction force} = \text{Applied force} = 10 \text{ N}$$

37. Ans: (d)

Sol:

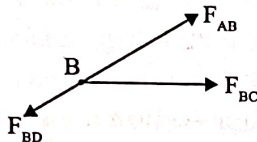


$$\Rightarrow V_R = (1R)\omega$$

$$\Rightarrow \omega = \frac{V_R}{(1R)} \Rightarrow \omega \times \frac{1}{\frac{1}{2}} \Rightarrow \omega = 2 V_R$$

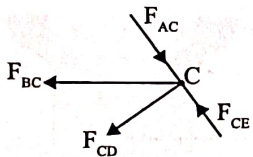
38. Ans: 0

Sol:



Adopting method of joints and taking FBD of joint B

$F_{BC} = 0$ (zero force member)



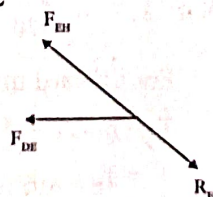
Further by taking FBD of joint C

$F_{CD} = 0$

39. Ans: (a)

Sol: If at any joint three forces are acting out of which two of them are collinear then force in third member must be zero.

- For member BF look at joint B.
- For member GC look at joint G.
- For member DH look at joint D.
- For member ED look at joint E

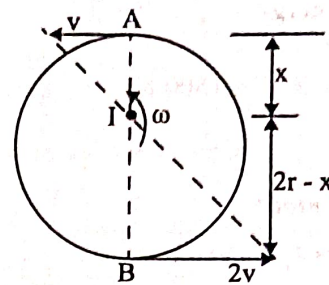


Since, F_{EH} & R_E are collinear

$\therefore F_{DE} = 0$

40. Ans: (d)

Sol:



$$\frac{v}{IA} = \frac{2v}{IB} = \omega$$

$$\frac{v}{x} = \frac{2v}{2r-x}$$

$$2r-x = 2x \Rightarrow x = \frac{2r}{3}$$

$$\omega = \frac{v}{x} = \frac{3v}{2r}$$

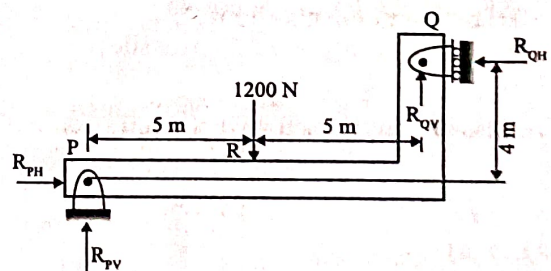
41. Ans: (d)

Sol: When the roller is about to climb the contact with floor is lost and hence normal reaction exerted by floor is zero.

Also when the roller is just about to climb the friction force will also be zero.

42. Ans: 1500

Sol:



Taking moments about point P

$$\Sigma m_P = 0$$

$$1200 \times 5 - R_{QH} \times 4 - R_{QV} \times 10 = 0$$

As the support at Q is roller support,

It doesn't restrict the motion in vertical direction.

$$\therefore R_{QV} = 0$$

$$\therefore 1200 \times 5 - R_{Q_H} \times 4 = 0$$

$$\therefore R_{Q_H} = 1500 \text{ N}$$

The resultant load on Q,

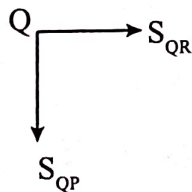
$$R_Q = \sqrt{R_{Q_H}^2 + R_{Q_V}^2} = 1500 \text{ N}$$

43. Ans: (a)

44. Ans: (d)

Sol:

(i) FBD of joint Q:



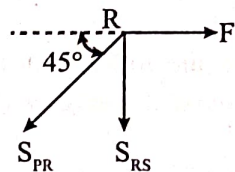
$$\Sigma F_x = 0 \Rightarrow S_{QR} = 0$$

$$\Sigma F_y = 0 \Rightarrow S_{QP} = 0$$

(ii) FBD of joint R:

i) $PQ = QR \therefore \angle QRP = 45^\circ$

ii) $S_{QP} = 0$



$$\Sigma F_x = 0 \Rightarrow F = S_{PR} \times \cos 45$$

$$\therefore S_{PR} = \sqrt{2} \cdot F \text{ \{tensile\}}$$

$$\Sigma F_y = 0 \Rightarrow S_{PR} \sin 45 + S_{RS} = 0$$

$$\therefore S_{RS} = -F = F \text{ \{compressive\}}$$

45. Ans: (a)

Sol: Screw efficiency

$$= \frac{\text{Work done in lifting the load/rev}}{\text{Work done by the applied force/rev}}$$

$$\text{Efficiency of screw jack } \eta = \frac{\tan \alpha}{\tan(\alpha + \phi)}$$

$\therefore$ Efficiency depends on helix angle and friction angle.

46. Ans: 10

Sol: Given: $I_{\text{rod}} = 0.1 \text{ kg.cm}^2 = 1 \times 10^{-5} \text{ kg.m}^2$

$$m_{\text{point}} = 10^{-3} \text{ kg}$$

$$V_{\text{point}} = 2 \text{ m/s}$$

$$\omega_o (\text{rod}) = 0 \text{ rad/s}$$

Plastic impact

$$\omega_{\text{final}} = ?$$

Since there is no external moment involved about O. Therefore the Angular momentum of the system about O is conserved.

$$I_{\text{rod}} \omega_{\text{initial}} + (m V_o r)_{\text{point}} = (I_{\text{rod}} + I_{\text{point}}) \omega_{\text{final}}$$

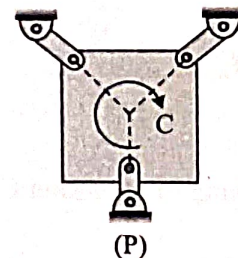
$$10^{-3} \times 2 \times 0.1 = (10^{-5} + 10^{-3} (10^{-1})^2) \omega_{\text{final}}$$

$$2 \times 10^{-4} = 2 \times 10^{-5} \omega_{\text{final}}$$

$$\Rightarrow \omega_{\text{final}} = 10 \text{ rad/s}$$

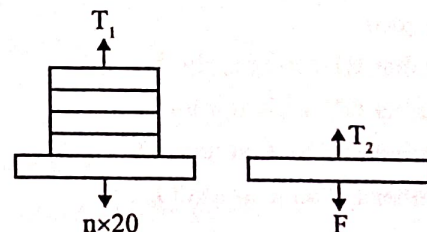
47. Ans: (b, c, d)

Sol: In case P it forms collinear force system. i.e., all the forces pass through the intersection point. Hence, they can't balance couple (As they can't create couple).



48. Ans: 3

Sol: F.B.D of fig.(i) :



For upward impending motion of left side platform

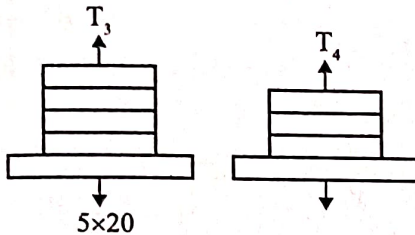
$$T_1 = n \times 20 = 5 \times 20 = 100 \text{ N} \text{ ---- (i)}$$

$$F = T_2 = 200 \text{ N} \text{ ---- (ii)}$$

$$\therefore \frac{T_2}{T_1} = 2$$

So, ratio of tensions on two sides remains [in figure (i) and (ii)] same as 2 because it depends on μ & θ which is same in fig.(i) and fig.(ii).

F.B.D of Fig (ii) :



$$T_3 = 100$$

Now since impending motion of left side platform is downward therefore T_3 is tight side tension.

$$T_4 = \frac{T_3}{2} = \frac{100}{2} = 50 \text{ N}$$

$$T_4 = m \times 20 \Rightarrow 50 = m \times 20$$

$$\Rightarrow m = 2.5$$

Hence, $m = 3$ (As no. of discs is integer)

49. Ans: 0.34

Sol: Density, $\rho = 7800 \text{ kg/m}^3$

Moment of inertia, $I = 4.36 \text{ kg-m}^2$

$$I = \frac{mR^2}{2} = \frac{\rho \pi R^2 t \times R^2}{2}$$

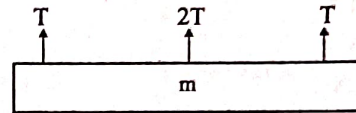
$$4.36 = \frac{7800 \times \pi R^4 \times 0.025}{2}$$

$$R = 0.345 \text{ meters}$$

Two Marks Solutions

01. Ans: (c)

Sol:



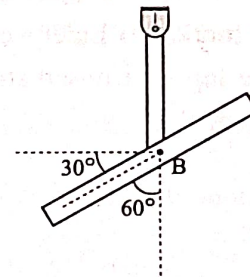
$$T + 2T + T = mg$$

$$4T = mg$$

$$m = 4T/g$$

02. Ans: (a)

Sol:



As centre of gravity passes through 'B'. There is no moment about 'B' so always the angle between two bars is constant.

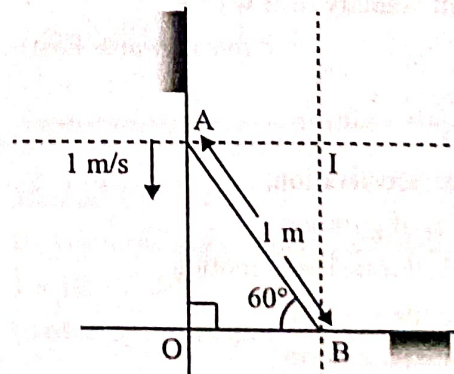
03. Ans: (a)

Sol: $V_a = 1 \text{ m/s}$

V_a = along vertical ,

V_b = along horizontal

So instantaneous center of V_a and V_b will be perpendicular to A and B respectively



$$IA = OB = l \times \cos \theta = 1 \times \cos 60^\circ = \frac{1}{2} \text{ m}$$

$$IB = OA = l \times \sin \theta = 1 \times \sin 60^\circ = \frac{\sqrt{3}}{2} \text{ m}$$

$$V_b = \omega \times IA$$

$$\Rightarrow \omega = \frac{V_b}{IA} = 2 \text{ rad/sec}$$

04. Ans : a - 4, b - 3, c - 1, d - 5

Sol:

- Collision of particles → Impulse-momentum principle
- Stability → Minimum potential energy
- Satellite motion → Euler's equation of motion
- Spinning top → Conservation of moment of momentum

05. Ans: (d)

Sol: $\tan \theta = \frac{125}{275}$

$$\Rightarrow \theta = 24.45^\circ$$

$$T \sin \theta = mg$$

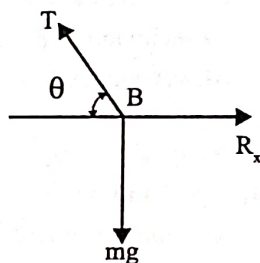
$$T \sin 24.45 = (35 \times 9.81)$$

$$\Rightarrow T = 829.5 \text{ N}$$

$$R_x = T \cos 24.45$$

$$= 755.4 \text{ N}$$

$$R_y = 0$$



06. Ans: (d)

Sol: Apparent velocity of B w.r.t A

$$= 8 \text{ m/s (towards East)}$$

07. Ans: (a)

Sol: a = linear acceleration,

k = radius of gyration

For vertical translation motion

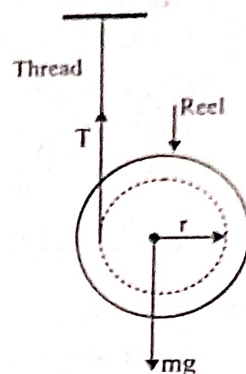
$$mg - T = ma \text{ ----- (1)}$$

For rotational motion

$$T \times r = I\alpha$$

$$Tr = mk^2 \alpha$$

$$= mk^2 \times \frac{a}{r}$$



$$\Rightarrow T = \frac{mk^2}{r^2} \times a \text{ ----- (2)}$$

$$mg - \frac{mk^2}{r^2} \times a = ma \Rightarrow a = \frac{gr^2}{(k^2 + r^2)}$$

08. Ans: (c)

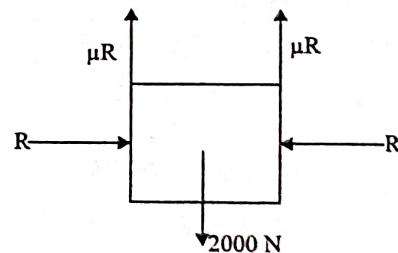
Sol: Tension in thread,

$$T = mg - ma$$

$$= mg - \frac{mr^2 g}{k^2 + r^2} = \frac{mgk^2}{r^2 + k^2}$$

09. Ans: (d)

Sol:



At equilibrium

$$2\mu R = 2000 \Rightarrow R = \frac{2000}{2 \times 0.1} = 10,000 \text{ N}$$

Taking moment about pin

$$10,000 \times 150 = F \times 300$$

$$\Rightarrow F = 5000 \text{ N}$$

10. Ans: (b)

Sol: $k = 981 \times 10^3 \text{ N/m}$, $\delta = 100 \text{ mm}$

$$mgh = \frac{1}{2} kx^2$$

$$h = \frac{1}{2} \times \frac{981 \times 10^3 \times (0.1)^2}{100 \times 9.81}$$

$$= 5 \text{ m} = 5000 \text{ mm}$$

11. Ans: (b)

Sol: $I_1 = 10 \times 0.2^2 = 0.4 \text{ kg-m}^2$

$$I_2 = 10 \times 0.2^2 + 20 \times 0.1^2 = 0.4 + 0.2 = 0.6 \text{ kg-m}^2$$

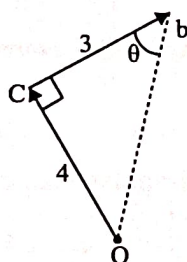
Percentage of increase in mass moment of inertia,

$$I = \frac{0.6 - 0.4}{0.4} \times 100 = 50\%$$

12. Ans: (c)

Sol: Velocity of bomb (V_b) = Velocity of canon
 $= \bar{V}_{\text{canon}} + \bar{V}_{bc}$

$$V_{\text{canon}} = r\omega = 2 \times 2 = 4 \text{ m/s}$$



$$V_b = \sqrt{3^2 + 4^2} = 5 \text{ m/s}$$

$$\tan \theta = \frac{4}{3}$$

$$\Rightarrow \theta = 53.13$$

13. Ans: (c)

Sol: $T_1 = mg - m\ddot{x}$

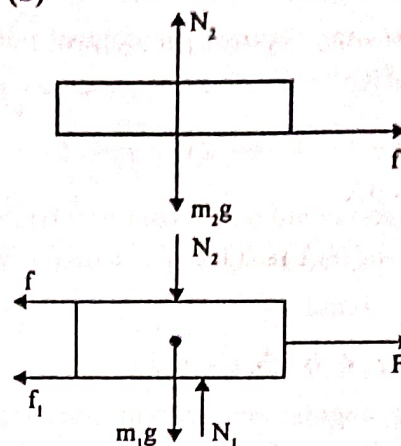
$$T_2 = mg + m\ddot{x}$$

$$\text{Power} = \frac{\int (T_2 - T_1) dx}{t}$$

$$= \frac{2m \int \ddot{x} dx}{t} = \frac{2m}{t} \int_0^2 v dv = \frac{mV^2}{t}$$

14. Ans: (b)

Sol:



where, f is the friction between the two books.

f_1 is the friction between the lower book and ground.

Now, maximum possible acceleration of upper book.

$$a_{\text{max}} = \frac{f_{\text{max}}}{m_2} = \frac{\mu m_2 g}{m_2} = \mu \times g = 0.3 \times 9.81 = 2.943 \text{ m/s}^2$$

For slip to occur, acceleration (a_1) of lower book.

$$\text{i.e., } a_1 \geq a_{\text{max}}$$

$$\frac{F - f - f_1}{m_1} \geq 2.943$$

$$F - 2.943 - 0.3 \times 2 \times 9.81 \geq 2.943$$

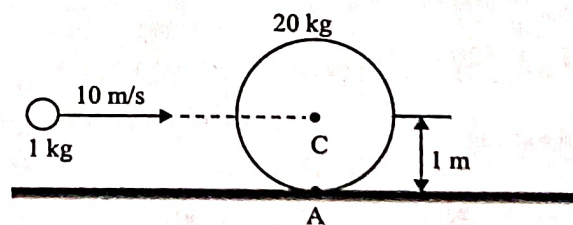
$$[\because f = f_{\text{max}} = 2.943 \text{ and } f_1 = \mu \times (m_1 + m_2)g = 0.3 \times 2 \times 9.81]$$

$$F \geq 11.77 \text{ N}$$

$$F_{\text{min}} = 11.77 \text{ N}$$

15. Ans: (a)

Sol:



Method I:

By conservation of linear momentum, we get

$$1 \times 10 = (20 + 1) \times V_{\text{cm}}$$

(where V_{cm} = velocity of centre of mass)

$$\Rightarrow V_{\text{cm}} = \frac{10}{21} \text{ m/s}$$

Applying angular momentum conservation about an axis passing through the contact point (A) and perpendicular to the plane of paper, we get

$$1 \times 10 \times 1 = I_{cm} \omega + 21 \times \frac{10}{21} \times 1$$

[Angular momentum about any axis passing through A can be written as, $\vec{L}_A = \vec{L}_{cm} + m(\vec{r} \times \vec{V}_{cm})$
 $\Rightarrow \omega = 0 \text{ rad/sec}$

Method II :

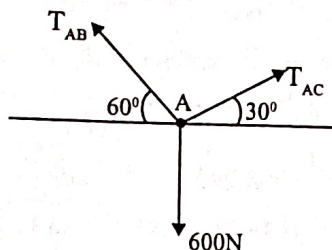
Applying angular momentum conservation about an axis passing through centre of wheel and perpendicular to the plane of paper,

$$\therefore 0 = I_{cm} \omega$$

$$\Rightarrow \omega = 0 \text{ rad/sec}$$

16. Ans: (a)

Sol:



$$T_{AB} \cos 60^\circ = T_{AC} \cos 30^\circ$$

$$T_{AB} = \sqrt{3} T_{AC}$$

$$T_{AB} \sin 60^\circ + T_{AC} \sin 30^\circ = 600 \text{ N}$$

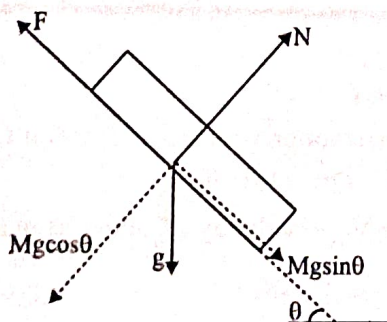
$$\frac{3}{2} T_{AC} + \frac{1}{2} T_{AC} = 600$$

$$T_{AC} = 300 \text{ N}$$

$$T_{AB} = 520 \text{ N}$$

17. Ans: (a)

Sol:



$$mg \sin \theta - F = ma$$

$$mg \cos \theta = N$$

$$F = \mu N$$

$$\Rightarrow mg \sin \theta - \mu N = ma$$

$$\Rightarrow mg \sin \theta - \mu mg \cos \theta = ma$$

$$\Rightarrow a = g(\sin \theta - \mu \cos \theta)$$

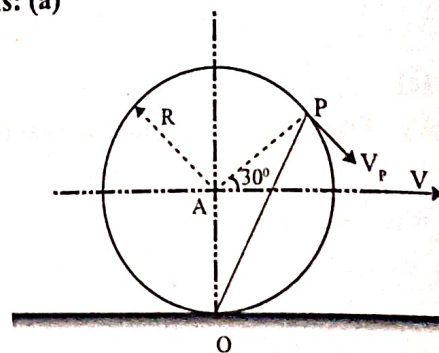
$$\text{Again } S = ut + \frac{1}{2} at^2 \Rightarrow S = \frac{1}{2} at^2 \quad (\because u = 0)$$

$$\Rightarrow t = \sqrt{\frac{2S}{a}} = \sqrt{\frac{2S}{g(\sin \theta - \mu \cos \theta)}}$$

$$\Rightarrow t = \sqrt{\frac{2S}{g \cos \theta (\tan \theta - \mu)}}$$

18. Ans: (a)

Sol:



Here 'O' is the instantaneous centre

$$V_P = \omega \times OP$$

$$V_A = R\omega$$

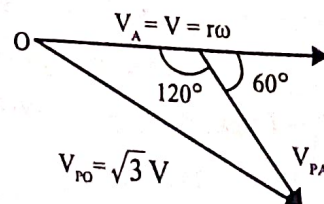
$$\text{In } \triangle OAP, \cos 120^\circ = \frac{R^2 + R^2 - OP^2}{2R \times R}$$

$$-0.5 = \frac{2R^2 - OP^2}{2R^2}$$

$$OP = \sqrt{3} R$$

$$V_P = \sqrt{3} R \times \omega = \sqrt{3} V$$

or



$$V_P = \vec{V}_A + \vec{V}_{PO} = \vec{V} + \vec{AP} \times \omega$$

$$= \sqrt{V^2 + V^2 + 2V^2 \cos 60^\circ} = \sqrt{3} V$$